140. Marcella has 6 fewer male cousins than female cousins. Let f represent the number of female cousins. Write an expression for the number of boy cousins.

141. Greg has nickels and pennies in his pocket. The number of pennies is seven less than twice the number of nickels. Let n represent the number of nickels. Write an expression for the number of pennies.

142. Jeannette has $\$5$ and $\$10$ bills in her wallet. The number of fives is three more than six times the number of tens. Let t represent the number of tens. Write an expression for the number of fives.

Everyday Math

In the following exercises, use algebraic expressions to solve the problem.

143. **Car insurance** Justin's car insurance has a $\$750$ deductible per incident. This means that he pays $\$750$ and his insurance company will pay all costs beyond $\$750$. If Justin files a claim for $\$2{,}100$, how much will he pay, and how much will his insurance company pay?

144. **Home insurance** Pam and Armando's home insurance has a $\$2{,}500$ deductible per incident. This means that they pay $\$2{,}500$ and their insurance company will pay all costs beyond $\$2{,}500$. If Pam and Armando file a claim for $\$19{,}400$, how much will they pay, and how much will their insurance company pay?

Writing Exercises

145. Explain why "the sum of *x* and *y*" is the same as "the sum of *y* and *x*," but "the difference of *x* and *y*" is not the same as "the difference of *y* and *x*." Try substituting two random numbers for x and y to help you explain.

146. Explain the difference between "4 times the sum of x and y" and "the sum of 4 times x and y."

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
evaluate algebraic expressions.			
identify terms, coefficients, and like terms.			
simplify expressions by combining like terms.			
translate word phrases to algebraic expressions.			

ⓑ *After reviewing this checklist, what will you do to become confident for all objectives?*

2.3 Solving Equations Using the Subtraction and Addition Properties of Equality

Learning Objectives

By the end of this section, you will be able to:

- Determine whether a number is a solution of an equation
- Model the Subtraction Property of Equality
- Solve equations using the Subtraction Property of Equality
- Solve equations using the Addition Property of Equality
- Translate word phrases to algebraic equations
- Translate to an equation and solve

☑ **BE PREPARED** 2.7 Before you get started, take this readiness quiz.

Evaluate $x + 8$ when $x = 11$.
If you missed this problem, review Example 2.13.

☑ **BE PREPARED** 2.8 Evaluate $5x - 3$ when $x = 9$.
If you missed this problem, review Example 2.14.

☑ **BE PREPARED** 2.9 Translate into algebra: the difference of x and 8.
If you missed this problem, review Example 2.24.

When some people hear the word *algebra*, they think of solving equations. The applications of solving equations are limitless and extend to all careers and fields. In this section, we will begin solving equations. We will start by solving basic equations, and then as we proceed through the course we will build up our skills to cover many different forms of equations.

Determine Whether a Number is a Solution of an Equation

Solving an equation is like discovering the answer to a puzzle. An algebraic equation states that two algebraic expressions are equal. To solve an equation is to determine the values of the variable that make the equation a true statement. Any number that makes the equation true is called a **solution** of the equation. It is the answer to the puzzle!

Solution of an Equation

A **solution to an equation** is a value of a variable that makes a true statement when substituted into the equation.

The process of finding the solution to an equation is called solving the equation.

To find the solution to an equation means to find the value of the variable that makes the equation true. Can you recognize the solution of $x + 2 = 7$? If you said 5, you're right! We say 5 is a solution to the equation $x + 2 = 7$ because when we substitute 5 for x the resulting statement is true.

$$x + 2 = 7$$
$$5 + 2 \stackrel{?}{=} 7$$
$$7 = 7 ✓$$

Since $5 + 2 = 7$ is a true statement, we know that 5 is indeed a solution to the equation.

The symbol $\stackrel{?}{=}$ asks whether the left side of the equation is equal to the right side. Once we know, we can change to an equal sign ($=$) or not-equal sign ($\neq$).

HOW TO

Determine whether a number is a solution to an equation.

Step 1. Substitute the number for the variable in the equation.
Step 2. Simplify the expressions on both sides of the equation.
Step 3. Determine whether the resulting equation is true.
- If it is true, the number is a solution.
- If it is not true, the number is not a solution.

EXAMPLE 2.28

Determine whether $x = 5$ is a solution of $6x - 17 = 16$.

Solution

	$6x - 17 = 16$
Substitute 5 for x.	$6 \cdot 5 - 17 \stackrel{?}{=} 16$
Multiply.	$30 - 17 \stackrel{?}{=} 16$
Subtract.	$13 \neq 16$

So $x = 5$ is not a solution to the equation $6x - 17 = 16$.

TRY IT 2.55 Is $x = 3$ a solution of $4x - 7 = 16$?

TRY IT 2.56 Is $x = 2$ a solution of $6x - 2 = 10$?

EXAMPLE 2.29

Determine whether $y = 2$ is a solution of $6y - 4 = 5y - 2$.

Solution

Here, the variable appears on both sides of the equation. We must substitute 2 for each y.

	$6y - 4 = 5y - 2$
Substitute 2 for y.	$6(2) - 4 \stackrel{?}{=} 5(2) - 2$
Multiply.	$12 - 4 \stackrel{?}{=} 10 - 2$
Subtract.	$8 = 8$ ✓

Since $y = 2$ results in a true equation, we know that 2 is a solution to the equation $6y - 4 = 5y - 2$.

TRY IT 2.57 Is $y = 3$ a solution of $9y - 2 = 8y + 1$?

TRY IT 2.58 Is $y = 4$ a solution of $5y - 3 = 3y + 5$?

Model the Subtraction Property of Equality

We will use a model to help you understand how the process of solving an equation is like solving a puzzle. An envelope represents the variable – since its contents are unknown – and each counter represents one.

Suppose a desk has an imaginary line dividing it in half. We place three counters and an envelope on the left side of desk, and eight counters on the right side of the desk as in Figure 2.3. Both sides of the desk have the same number of counters, but some counters are hidden in the envelope. Can you tell how many counters are in the envelope?

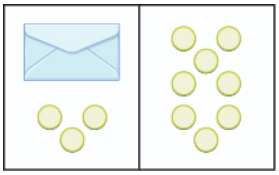


Figure 2.3

What steps are you taking in your mind to figure out how many counters are in the envelope? Perhaps you are thinking "I need to remove the 3 counters from the left side to get the envelope by itself. Those 3 counters on the left match with 3 on the right, so I can take them away from both sides. That leaves five counters on the right, so there must be 5 counters in the envelope." Figure 2.4 shows this process.

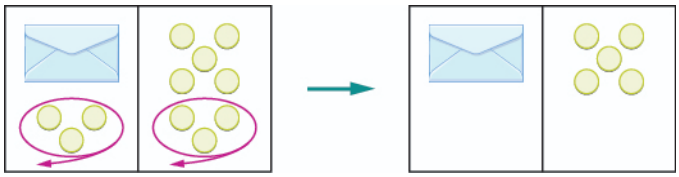


Figure 2.4

What algebraic equation is modeled by this situation? Each side of the desk represents an expression and the center line takes the place of the equal sign. We will call the contents of the envelope x, so the number of counters on the left side of the desk is $x + 3$. On the right side of the desk are 8 counters. We are told that $x + 3$ is equal to 8 so our equation is$x + 3 = 8$.

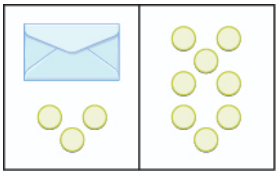


Figure 2.5

$x + 3 = 8$

Let's write algebraically the steps we took to discover how many counters were in the envelope.

	$x + 3 = 8$
First, we took away three from each side.	$x + 3 - 3 = 8 - 3$
Then we were left with five.	$x = 5$

Now let's check our solution. We substitute 5 for x in the original equation and see if we get a true statement.

$x + 3 = 8$

$5 + 3 \stackrel{?}{=} 8$

$8 = 8$ ✓

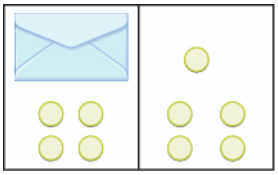
Our solution is correct. Five counters in the envelope plus three more equals eight.

MANIPULATIVE MATHEMATICS

Doing the Manipulative Mathematics activity, "Subtraction Property of Equality" will help you develop a better understanding of how to solve equations by using the Subtraction Property of Equality.

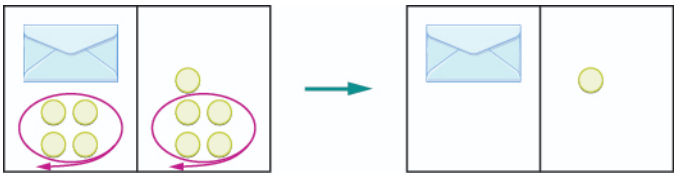
EXAMPLE 2.30

Write an equation modeled by the envelopes and counters, and then solve the equation:



Solution

On the left, write x for the contents of the envelope, add the 4 counters, so we have $x+4$.	$x+4$
On the right, there are 5 counters.	5
The two sides are equal.	$x+4=5$
Solve the equation by subtracting 4 counters from each side.	



We can see that there is one counter in the envelope. This can be shown algebraically as:

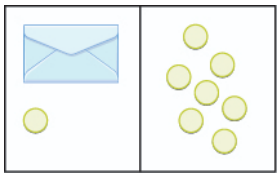
$$\begin{aligned} x+4&=5\\ x+4-4&=5-4\\ x&=1 \end{aligned}$$

Substitute 1 for x in the equation to check.

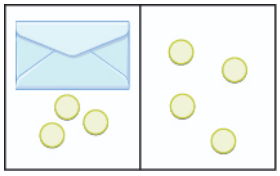
$$\begin{aligned} x+4&=5\\ 1+4&\stackrel{?}{=}5\\ 5&=5\ \checkmark \end{aligned}$$

Since $x=1$ makes the statement true, we know that 1 is indeed a solution.

TRY IT 2.59 Write the equation modeled by the envelopes and counters, and then solve the equation:



TRY IT 2.60 Write the equation modeled by the envelopes and counters, and then solve the equation:



Solve Equations Using the Subtraction Property of Equality

Our puzzle has given us an idea of what we need to do to solve an equation. The goal is to isolate the variable by itself on

one side of the equations. In the previous examples, we used the Subtraction Property of Equality, which states that when we subtract the same quantity from both sides of an equation, we still have equality.

Subtraction Property of Equality

For any numbers a, b, and c, if

$$a = b$$

then

$$a - c = b - c$$

Think about twin brothers Andy and Bobby. They are 17 years old. How old was Andy 3 years ago? He was 3 years less than 17, so his age was $17 - 3$, or 14. What about Bobby's age 3 years ago? Of course, he was 14 also. Their ages are equal now, and subtracting the same quantity from both of them resulted in equal ages 3 years ago.

$$a = b$$
$$a - 3 = b - 3$$



HOW TO

Solve an equation using the Subtraction Property of Equality.

Step 1. Use the Subtraction Property of Equality to isolate the variable.
Step 2. Simplify the expressions on both sides of the equation.
Step 3. Check the solution.

EXAMPLE 2.31

Solve: $x + 8 = 17$.

Solution

We will use the Subtraction Property of Equality to isolate x.

	$x + 8 = 17$
Subtract 8 from both sides.	$x + 8 - 8 = 17 - 8$
Simplify.	$x = 9$
	$x + 8 = 17$
	$9 + 8 = 17$
	$17 = 17$ ✓

Since $x = 9$ makes $x + 8 = 17$ a true statement, we know 9 is the solution to the equation.

TRY IT 2.61 Solve:

$$x + 6 = 19$$

TRY IT 2.62 Solve:

$$x + 9 = 14$$

EXAMPLE 2.32

Solve: $100 = y + 74$.

Solution

To solve an equation, we must always isolate the variable—it doesn't matter which side it is on. To isolate y, we will subtract 74 from both sides.

	$100 = y + 74$
Subtract 74 from both sides.	$100 - 74 = y + 74 - 74$
Simplify.	$26 = y$
Substitute 26 for y to check. $100 = y + 74$ $100 \stackrel{?}{=} 26 + 74$ $100 = 100$ ✓	

Since $y = 26$ makes $100 = y + 74$ a true statement, we have found the solution to this equation.

TRY IT 2.63 Solve:

$$95 = y + 67$$

TRY IT 2.64 Solve:

$$91 = y + 45$$

Solve Equations Using the Addition Property of Equality

In all the equations we have solved so far, a number was added to the variable on one side of the equation. We used subtraction to "undo" the addition in order to isolate the variable.

But suppose we have an equation with a number subtracted from the variable, such as $x - 5 = 8$. We want to isolate the variable, so to "undo" the subtraction we will add the number to both sides.

We use the Addition Property of Equality, which says we can add the same number to both sides of the equation without changing the equality. Notice how it mirrors the Subtraction Property of Equality.

Addition Property of Equality

For any numbers a, b, and c, if

$$a = b$$

then

$$a + c = b + c$$

Remember the 17-year-old twins, Andy and Bobby? In ten years, Andy's age will still equal Bobby's age. They will both be 27.

$$a = b$$
$$a + 10 = b + 10$$

We can add the same number to both sides and still keep the equality.

HOW TO

Solve an equation using the Addition Property of Equality.

Step 1. Use the Addition Property of Equality to isolate the variable.
Step 2. Simplify the expressions on both sides of the equation.
Step 3. Check the solution.

EXAMPLE 2.33

Solve: $x - 5 = 8$.

Solution

We will use the Addition Property of Equality to isolate the variable.

	$x - 5 = 8$
Add 5 to both sides.	$x - 5 + 5 = 8 + 5$
Simplify.	$x = 13$
Now we can check. Let $x = 13$.	
$x - 5 = 8$	
$13 - 5 \stackrel{?}{=} 8$	
$8 = 8$ ✓	

TRY IT 2.65 Solve:

$x - 9 = 13$

TRY IT 2.66 Solve:

$y - 1 = 3$

EXAMPLE 2.34

Solve: $27 = a - 16$.

Solution

We will add 16 to each side to isolate the variable.

	$27 = a - 16$
Add 16 to each side.	$27 + 16 = a - 16 + 16$
Simplify.	$43 = a$
Now we can check. Let $a = 43$.	$27 = a - 16$
	$27 \stackrel{?}{=} 43 - 16$
	$27 = 27$ ✓

The solution to $27 = a - 16$ is $a = 43$.

TRY IT 2.67 Solve:

$19 = a - 18$

TRY IT 2.68 Solve:

$27 = n - 14$

Translate Word Phrases to Algebraic Equations

Remember, an equation has an equal sign between two algebraic expressions. So if we have a sentence that tells us that two phrases are equal, we can translate it into an equation. We look for clue words that mean *equals.* Some words that translate to the equal sign are:

- is equal to
- is the same as
- is
- gives
- was
- will be

It may be helpful to put a box around the *equals* word(s) in the sentence to help you focus separately on each phrase. Then translate each phrase into an expression, and write them on each side of the equal sign.

We will practice translating word sentences into algebraic equations. Some of the sentences will be basic number facts with no variables to solve for. Some sentences will translate into equations with variables. The focus right now is just to translate the words into algebra.

EXAMPLE 2.35

Translate the sentence into an algebraic equation: The sum of 6 and 9 is 15.

Solution

The word *is* tells us the equal sign goes between 9 and 15.

Locate the "equals" word(s).	The sum of 6 and 9 is 15.
Write the = sign.	The sum of 6 and 9 = 15.

Translate the words to the left of the *equals* word into an algebraic expression.	$6 + 9 = ___$
Translate the words to the right of the *equals* word into an algebraic expression.	$6 + 9 = 15$

TRY IT 2.69 Translate the sentence into an algebraic equation:

The sum of 7 and 6 gives 13.

TRY IT 2.70 Translate the sentence into an algebraic equation:

The sum of 8 and 6 is 14.

EXAMPLE 2.36

Translate the sentence into an algebraic equation: The product of 8 and 7 is 56.

Solution

The location of the word *is* tells us that the equal sign goes between 7 and 56.

Locate the "equals" word(s).	The product of 8 and 7 is 56.
Write the = sign.	The product of 8 and 7 = 56.
Translate the words to the left of the *equals* word into an algebraic expression.	$8 \cdot 7 = ___$
Translate the words to the right of the *equals* word into an algebraic expression.	$8 \cdot 7 = 56$

TRY IT 2.71 Translate the sentence into an algebraic equation:

The product of 6 and 9 is 54.

TRY IT 2.72 Translate the sentence into an algebraic equation:

The product of 21 and 3 gives 63.

EXAMPLE 2.37

Translate the sentence into an algebraic equation: Twice the difference of x and 3 gives 18.

Solution

Locate the "equals" word(s).	Twice the difference of x and 3 gives 18.
Recognize the key words: *twice; difference of .... and ....*	*Twice* means two times.
Translate.	Twice the difference of x and 3 gives 18. $2 \quad (x - 3) \quad = \quad 18$

TRY IT 2.73 Translate the given sentence into an algebraic equation:

Twice the difference of x and 5 gives 30.

TRY IT 2.74 Translate the given sentence into an algebraic equation:

Twice the difference of y and 4 gives 16.

Translate to an Equation and Solve

Now let's practice translating sentences into algebraic equations and then solving them. We will solve the equations by using the Subtraction and Addition Properties of Equality.

EXAMPLE 2.38

Translate and solve: Three more than x is equal to 47.

Solution

	Three more than x is equal to 47.
Translate.	$x+3=47$
Subtract 3 from both sides of the equation.	$x+3-3=47-3$
Simplify.	$x=44$
We can check. Let $x=44$.	$x+3=47$
	$44+3\stackrel{?}{=}47$
	$47=47$ ✓

So $x=44$ is the solution.

TRY IT 2.75 Translate and solve:

Seven more than x is equal to 37.

TRY IT 2.76 Translate and solve:

Eleven more than y is equal to 28.

EXAMPLE 2.39

Translate and solve: The difference of y and 14 is 18.

Solution

	The difference of y and 14 is 18.
Translate.	$y-14=18$

Add 14 to both sides.		$y - 14 + 14 = 18 + 14$
Simplify.		$y = 32$
We can check. Let $y = 32$.	$y - 14 = 18$	
	$32 - 14 \stackrel{?}{=} 18$	
	$18 = 18$ ✓	

So $y = 32$ is the solution.

TRY IT 2.77 Translate and solve:

The difference of z and 17 is equal to 37.

TRY IT 2.78 Translate and solve:

The difference of x and 19 is equal to 45.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Solving One Step Equations By Addition and Subtraction (https://openstaxcollege.org/l/24Solveonestep)

SECTION 2.3 EXERCISES

Practice Makes Perfect

Determine Whether a Number is a Solution of an Equation

In the following exercises, determine whether each given value is a solution to the equation.

147. $x + 13 = 21$
ⓐ $x = 8$ ⓑ $x = 34$

148. $y + 18 = 25$
ⓐ $y = 7$ ⓑ $y = 43$

149. $m - 4 = 13$
ⓐ $m = 9$ ⓑ $m = 17$

150. $n - 9 = 6$
ⓐ $n = 3$ ⓑ $n = 15$

151. $3p + 6 = 15$
ⓐ $p = 3$ ⓑ $p = 7$

152. $8q + 4 = 20$
ⓐ $q = 2$ ⓑ $q = 3$

153. $18d - 9 = 27$
ⓐ $d = 1$ ⓑ $d = 2$

154. $24f - 12 = 60$
ⓐ $f = 2$ ⓑ $f = 3$

155. $8u - 4 = 4u + 40$
ⓐ $u = 3$ ⓑ $u = 11$

156. $7v - 3 = 4v + 36$
ⓐ $v = 3$ ⓑ $v = 11$

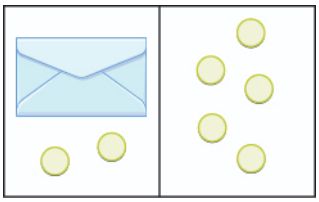
157. $20h - 5 = 15h + 35$
ⓐ $h = 6$ ⓑ $h = 8$

158. $18k - 3 = 12k + 33$
ⓐ $k = 1$ ⓑ $k = 6$

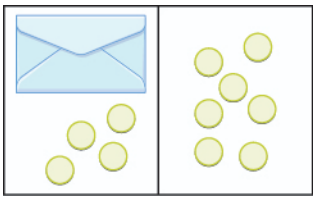
Model the Subtraction Property of Equality

In the following exercises, write the equation modeled by the envelopes and counters and then solve using the subtraction property of equality.

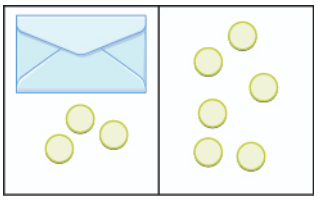
159.



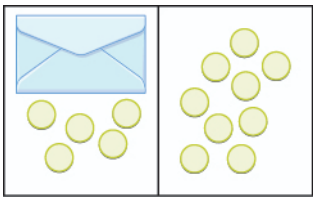
160.



161.



162.



Solve Equations using the Subtraction Property of Equality

In the following exercises, solve each equation using the subtraction property of equality.

163. $a + 2 = 18$

164. $b + 5 = 13$

165. $p + 18 = 23$

166. $q + 14 = 31$

167. $r + 76 = 100$

168. $s + 62 = 95$

169. $16 = x + 9$

170. $17 = y + 6$

171. $93 = p + 24$

172. $116 = q + 79$

173. $465 = d + 398$

174. $932 = c + 641$

Solve Equations using the Addition Property of Equality

In the following exercises, solve each equation using the addition property of equality.

175. $y - 3 = 19$

176. $x - 4 = 12$

177. $u - 6 = 24$

178. $v - 7 = 35$

179. $f - 55 = 123$

180. $g - 39 = 117$

181. $19 = n - 13$

182. $18 = m - 15$

183. $10 = p - 38$

184. $18 = q - 72$

185. $268 = y - 199$

186. $204 = z - 149$

Translate Word Phrase to Algebraic Equations

In the following exercises, translate the given sentence into an algebraic equation.

187. The sum of 8 and 9 is equal to 17.

188. The sum of 7 and 9 is equal to 16.

189. The difference of 23 and 19 is equal to 4.

190. The difference of 29 and 12 is equal to 17.

191. The product of 3 and 9 is equal to 27.

192. The product of 6 and 8 is equal to 48.

193. The quotient of 54 and 6 is equal to 9.

194. The quotient of 42 and 7 is equal to 6.

195. Twice the difference of n and 10 gives 52.

196. Twice the difference of m and 14 gives 64.

197. The sum of three times y and 10 is 100.

198. The sum of eight times x and 4 is 68.

Translate to an Equation and Solve

In the following exercises, translate the given sentence into an algebraic equation and then solve it.

199. Five more than p is equal to 21.

200. Nine more than q is equal to 40.

201. The sum of r and 18 is 73.